

Nicole Hee-Yeon Kim

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Nationality: United States of America and Republic of Korea (Dual Citizenship)

RESEARCH INTEREST

Efficient and Scalable AI Systems; Optimization for Large-Scale Deep Learning Models; Training and Inference Efficiency in LLMs and VLMs; Robust and Reliable AI

EDUCATION

- **Korea Advanced Institute of Science and Technology** Feb 2024 - Feb 2026
Master's Program, Department of Industrial & Systems Engineering, Overall GPA: 3.85/4.0 Daejeon, South Korea
- **Yonsei University** Mar 2019 - Feb 2024
Bachelor's Program, Department of Industrial Engineering, Overall GPA: 3.70/4.0 Seoul, South Korea

PUBLICATIONS

1. Lee, Y.*, Shin, J.*, **Kim, N.***, Bang, J., Lee, J., Hwang, K., Porikli, F., Song, H. (2026). Rethinking RAG in Long Videos: What to Retrieve and How to Use It?. *Submitted to NeurIPS 2026* (* Equal contribution.)
2. **Kim, N.** and Song, H. (2025). Robust Dataset Condensation using Supervised Contrastive Learning. In *Proceedings of International Conference on Computer Vision (ICCV 2025)*, Main Conference Paper.
3. **Kim, N.**, Choi, J., Lee, Y., Song, H. (2025). Robust Dataset Condensation via Semi-Supervised Learning. In *Proceedings of the Korea Computer Congress (KCC 2025)*, Selected for Oral Presentation.
4. **Kim, N.**, Lee, Y., Song, H. (2024). Robust Dataset Condensation via Supervised Contrastive Learning. In *Proceedings of the Korea Software Congress (KSC 2024)*, Selected for Oral Presentation.
5. Ban, M., Choi, J., Min, H., **Kim, N.**, Kim, M., Lee, J., Song, H. Completing Missing Annotation: Multi-Agent Debate for Accurate and Scalable Relevant Assessment for IR Benchmarks. In *Proceedings of International Conference on Learning Representations (ICLR 2026)*.
6. Lee, Y., Deng, J., **Kim, N.**, Min, H., Yun, T., Ban, M., Song, H. (2025). Towards a Holistic and Automated Evaluation Framework for Multi-Level Comprehension of LLMs in Book-Length Contexts. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*, Main Conference Paper.
7. Min, H., Lee, Y., Ban, M., Deng, J., **Kim, N.**, Yun, T., Su, H., Cai, J., Song, H. (2025). Towards Multi-dimensional Evaluation of LLM Summarization across Domains and Languages. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, Main Conference Paper.
8. Kim, T., Lee, S., Kang, J., Choi, Y., Yun, W., **Kim, N.**, Chen, Z., Xie, L., Song, K. (2025). IMC: A Benchmark for Invariant Learning under Multiple Causes. In *Proceedings of the CVPR 2025 Workshop on Domain Generalization: Evolution, Breakthroughs, and Future Horizons*, Best Paper Award.
9. Oh, J., Choi, J., **Kim, N.**, Yun, T., Kwon, R., Song, H. (2025). Learning to Verify Summary Facts with Fine-Grained LLM Feedback. In *Proceedings of the 2025 International Conference on Computational Linguistics (COLING 2025)*, Selected for Oral Presentation.
10. Oh, J., Choi, J., **Kim, N.**, Song, H. (2025). Improving Language Model Quality through LLM-based Fine-Grained Hallucinated Summary Generation. *Journal of Computing Practice*, vol. 31(2), pp. 91-97.
11. Oh, J., Choi, J., **Kim, N.**, Song, H. (2024). Improving the Text Summary Quality Through Understanding the Hallucination Level of Summarization Using Large Language Models. In *Proceedings of the Korea Computer Congress (KCC 2024)*, Selected for Oral Presentation.

RESEARCH EXPERIENCE

- **DISLab, Department of Industrial & Systems Engineering, KAIST** Feb 2024 - Feb 2026
Master's Student Daejeon, South Korea
 - Led the research project “Robust Dataset Condensation using Supervised Contrastive Learning” as the sole first author (accepted at ICCV 2025).
 - Led the research project “Qualcomm Industry–Academia Collaboration Project” titled “Rethinking RAG in Long Videos: What to Retrieve and How to Use It?”.
 - Participated in research project “Development of VLM-based PDF Document Understanding and Automated Road Alignment Coordinate Extraction Technology”, supported by ETRI.
- **MLAI Lab, Department of Applied Statistics, Yonsei University** Sep 2023 - Feb 2024
Undergraduate Research Intern Seoul, South Korea
 - Designed and constructed the PlayingCard-10 dataset, a core component of "IMC: A Benchmark for Invariant Learning under Multiple Causes" (CVPR 2025 Workshop Best Paper Award).

PATENT

- Robust Dataset Condensation using Supervised Contrastive Learning (2025, Pending).
- Cost-efficient and Accurate Debate-based Relevance Assessment System with Multi-agents(2025, Pending).

SCHOLARSHIPS

- **KAIST Support Scholarship** Feb 2024 - Feb 2026
Government-funded full tuition scholarship for M.S. program KAIST
- **Brain Korea 21 (BK21) Scholarship** Feb 2024 - Feb 2026
Government-funded research scholarship for graduate students National Research Foundation of Korea
- **Yonsei Welfare Scholarship** Mar 2019 - Jun 2023
Full tuition scholarship for B.S. program Yonsei University
- **University Innovation Support Scholarship** Sep 2020 - Feb 2021
Scholarship awarded for the development and advancement of an innovative start-up idea Yonsei University

HONORS AND AWARDS

- **Best Paper Award** Jun 2025
CVPR 2025 Workshop on Domain Generalization: Evolution, Breakthroughs, and Future Horizons
- **Outstanding Presentation Paper Award** Jul 2024
Korea Computer Congress 2024
- **1st Place, Promotional Video Contest** Feb 2022
Department of Industrial Engineering, Yonsei University
- **Yonsei Social Entrepreneurship Award** Feb 2021
Yonsei University

LANGUAGE PROFICIENCY

- **Languages:** Korean (Native), English (Fluent, TOEFL iBT: 111 [R30 L27 S28 W26])

TEACHING EXPERIENCE

- **Yonsei–NAVER Cloud Data Science Program** Dec 2020 - Dec 2023
Assisted in designing, editing, and organizing instructional materials and coding sessions Yonsei University

LEADERSHIP EXPERIENCE

- **Academic & Campus Life Mentor for Freshmen** Oct 2019 - Aug 2020
Institute for Higher Education Innovation, Yonsei University
- **Freshman Class President** Mar 2019 - Aug 2019
Department of Industrial Engineering, Yonsei University